

Assessment Task 3: Professional Learning Report

Name: Zhonghe Wang

Student ID: 25744879

Portfolio: <https://bellelynn.github.io/uts-43030-portfolio>

1. Introduction

This report reflects on the professional development I have undertaken in fulfilment of the Professional Learning Plan I submitted in Assessment Task 1. The PLP identified four learning goals worth roughly 200 effort hours in total; this report concentrates on the three I have made measurable progress against this semester (Goals 1, 2, and 4), which together account for the 150 hours and 6 credit points the Report requires. The fourth — Software Development Tools — is discussed briefly in Section 6 as ongoing work.

Some context on where I am coming from. My undergraduate background is a Bachelor of Laws, and I have spent the past two years deliberately retraining into software engineering. That trajectory shapes the report. It is the reason Goal 1 (ethics and Indigenous data sovereignty) hit harder than I expected — I came in with legal vocabulary but no architectural instinct — and it is the reason I have been so focused this semester on the technical breadth (full-stack JavaScript, network-layer security) that the graduate ads at my target employers actually ask for. Those employers, as set out in the PLP, are Fire Front Solutions (Junior / Graduate Mobile App Developer, my entry-level goal) and Holocentric (Software Developer Fullstack, my mid-term goal). Both build mission-critical Australian software where privacy, security, and ethics decisions sit alongside delivery.

The structure of the report follows the PLP. Section 2 explains why I chose each activity, Section 3 presents what was produced, Section 4 indexes the supporting evidence in my portfolio, Section 5 is the reflection on what changed in my thinking, and Section 6 sets out three future goals — one of which carries the deferred PLP Goal 3 forward. Together these demonstrate progress against the graduate attributes the subject is built around: socially responsible, technically proficient, and collaborative and communicative (Australian Computer Society [ACS], 2022).

2. Justification of Professional Development Activities

The PLP identified four required and professional skill gaps. I have chosen to reflect on three of them in this report: Goal 1 (ethical principles and Indigenous data sovereignty), Goal 2 (foundational cybersecurity), and Goal 4 (front-end and full-stack frameworks). The fourth, PLP Goal 3 (Software Development Tools — Jira, CI/CD, Docker, NUnit), is still in active progress through my university group projects and is more honestly reported as a future goal in Section 6 than claimed as completed work here. Concentrating on three deeper activities was a deliberate choice over spreading the available hours thinly across four.

2.1 Activity 1 — Ethical principles and Indigenous data sovereignty (PLP Goal 1)

Holocentric develops enterprise Risk and Compliance platforms for Australian government and regulated industries. Reading through their case studies and recent job ads made it clear that engineers there are expected to treat privacy and data ethics as architectural concerns rather than post-hoc legal review. My situation was unusual: my Law degree had given me strong general ethical reasoning and a working knowledge of privacy law, but I had never let any of it change a database schema. So I committed in parallel to finishing the UTS Canvas modules on professional ethics, the IT code of ethics, and Indigenous data sovereignty, and to using what I learned to audit the data model of LEXIS, my Law Firm Case Management System.

Pairing the theory with an applied audit was deliberate. Theory on its own, in my case, would have been the same legal vocabulary I had already arrived with.

2.2 Activity 2 — Foundational cybersecurity (PLP Goal 2)

My PLP scoped this goal around the OWASP Top 10 and a refactor of my Law Firm project's login/data entry forms. Three weeks in, I changed direction. Two informational interviews with UTS alumni working in security-adjacent roles, plus a closer reading of the Australian Cyber Security Centre's Essential Eight (ACSC, 2023), made it clear that without a network-layer foundation I would not really understand the threat model my code runs inside. The OWASP Top 10 alone would have given me a vocabulary but not an architecture. I enrolled in the Cisco Networking Academy's Network Security course instead and treated it as the foundation that the OWASP work would sit on top of later. Rescoping mid-semester was uncomfortable at the time, but being able to explain why I did it is itself the kind of strategic self-management the subject treats as a graduate attribute.

2.3 Activity 3 — Full-stack development on the JavaScript ecosystem (PLP Goal 4)

The PLP committed me to learning React (web) and React Native (mobile) on top of my existing C# and Python back-end skills, with the goal of becoming employable for the front-end-heavy roles at both target companies. Two things shifted during the semester. First, I realised that learning the JavaScript ecosystem end-to-end — front-end framework plus a Node-based back-end — was a more honest match for the actual job market than learning React on top of a C# back-end. The Holocentric Fullstack ad listed Node alongside React; Fire Front's stack is JavaScript-first. Second, I decided to prioritise depth on web (React + Vite + Express + PostgreSQL) over breadth across web and mobile. React Native is on the roadmap, but in 70 hours I could either build something credible on the web or build something shallow on both. I chose depth. The output was LEXIS, a complete full-stack web application built entirely on the JavaScript stack.

3. Evidence of Achievement

This section summarises what each activity produced. All artefacts — completion screenshots, audit documents, code, and the full LEXIS source repository — are hosted in the public portfolio linked on the title page, indexed in Section 4.

3.1 Ethical principles and Indigenous data sovereignty

I completed all the required UTS Canvas modules covering professional ethical principles, the ACS Code of Professional Conduct, the Australian Privacy Principles (Office of the Australian Information Commissioner [OAIC], 2022), and Indigenous data sovereignty (Maia Wingara Indigenous Data Sovereignty Collective, 2018). Beyond completion, I produced two concrete deliverables. First, I ran a privacy audit on the LEXIS PostgreSQL schema, classifying every personal-data field into three categories (client identifiers, matter details, billing data) and documenting the legal basis, retention period, and authorised role for each. Second, I refactored the data access layer so role-based access control (RBAC) is enforced inside the SQL queries — every read against client or case data now includes a tenant- and role-scoped WHERE clause, rather than relying on the route handler alone. The original design quietly allowed a class of accidental data exposure; the refactored one does not.

3.2 Foundational cybersecurity

I completed the majority of the Cisco Network Security course, including the modules on secure network design, access control, firewall technologies, and IPS. I deliberately did not sit the final certification exam — the practical lab content was worth doing properly, and partial completion with real understanding was a more honest outcome than a finished certificate I rushed through. The course has already changed how I write code. I now reason explicitly about which interfaces of the LEXIS API are network-exposed, and I have added defensive middleware to the Express back-end — body-size limits, content-type validation, JWT verification at the route boundary, and per-IP rate limiting — as a direct result.

3.3 Full-stack development on the JavaScript ecosystem

I shipped LEXIS as a complete single-page web application. The front-end uses React 19 with Vite as the build tool, Ant Design 6 for components, ECharts for the dashboard's case-growth chart, React Router DOM v7 for client-side routing, and Axios for HTTP. The back-end is a Node.js / Express 5 REST API talking to PostgreSQL via the pg driver, with auth handled by JWT bearer tokens issued at login. The database is normalised across five tables (Office, Lawyer, Client, Case, Case Assignment) with foreign-key constraints and a many-to-many junction for case-lawyer assignments. All four core workflows scoped in the PLP are working end-to-end: case listing with sort and filter, case detail with reassignment, client management, and a statistics dashboard. The complete source is public at github.com/bellelynn/case-management.

4. Integration with the PLP and Portfolio

This Report, my portfolio, and the original PLP are meant to be read as one consistent narrative. The PLP set the four goals and their success criteria. The portfolio holds the underlying artefacts — the audit document, the RBAC refactor, the Cisco progress evidence, and the full LEXIS source repository — organised under one section per goal reflected on in this Report, so any claim made here can be traced directly to its primary evidence. The table below provides the index.

PLP Goal	Portfolio Section	Key Artefacts
Goal 1 — Ethics & Indigenous Data Sovereignty	/ethics-privacy/	Canvas module screenshots; LEXIS privacy audit; RBAC enforcement in SQL queries; data classification table
Goal 2 — Cybersecurity	/cybersecurity/	Cisco Networking Academy progress dashboard; module-level completion screenshots; LEXIS Express middleware code; mid-semester rescope note
Goal 4 — Full-Stack (LEXIS)	/lexis/	Full source code at github.com/bellelynn/case-management ; architecture diagram; LEXIS UI screenshots; CaseList component code

Two things shifted from the original plan and I want to flag them rather than hide them. The cybersecurity goal was rescoped mid-semester from application-layer to network-layer

foundations, as justified in Section 2.2; the rescope note in the portfolio's cybersecurity section keeps that change on the record. The front-end goal also narrowed — from React + React Native to React on the web only — for the depth-over-breadth reasoning explained in Section 2.3.

5. Reflection on Progress and Learning

Schön's (1983) distinction between reflection-in-action and reflection-on-action gave me a useful frame for this section. Rather than retelling what happened, I have tried to be honest about what I now understand differently because of it.

5.1 What my Law degree gave me, and what it did not

Before this semester, I assumed my legal training was a complete advantage for the ethics work. It was a partial one. The Law degree gave me a strong instinct for legal-style reasoning — what does the statute say, what is the duty, who is the data subject — but it had not given me any sense that privacy is an architectural property of a system rather than a compliance overlay on top of one. Auditing my own LEXIS database forced that shift. Every schema decision turned out to be a privacy decision: what gets collected, how long it is kept, who can join which tables. The Indigenous data sovereignty module pushed it further by challenging an assumption I had not noticed I was making — that data, once collected, simply belongs to the collector. I now design data models by answering two questions up front: who has authority over this data, and what is the minimum collection that does the job. That is a different practice from the one I started the semester with.

5.2 Replanning is a professional skill, not a failure

I revised the scope of two of the three goals during the semester. Cybersecurity moved from application-layer OWASP to network-layer Cisco, and Goal 4 narrowed from React-plus-React-Native to React on the web only. Both decisions were uncomfortable at the time and both look right with hindsight. Honest progress against a revised plan is worth more than nominal completion of an unrealistic one. I expect to hit the same trade-off in industry — renegotiating scope without losing stakeholder trust is just part of the job — and being able to explain the rescoping on paper is itself the kind of evidence employers actually want.

5.3 The real cost of switching language ecosystems, and what I would do differently

Moving from C# / Python to a JavaScript-everywhere stack was the hardest part of the semester. It was not just learning React — it was unlearning the synchronous, strongly-typed, batteries-included mindset that ASP.NET Core had given me. The single worst offender was the `useEffect` dependency array: my instinct from C# component lifecycles was systematically wrong, and I burned several evenings debugging re-render loops before my mental model caught up. The takeaway is bigger than React. Real ecosystem fluency is not vocabulary — it is a rearrangement of how I think about state, time, and failure. With hindsight, two things I would do differently: start the JavaScript stack earlier, since I underestimated how much headspace the ecosystem shift would consume; and track hours against goals rather than just against the calendar, which would have made the mid-semester cybersecurity rescope happen two weeks sooner.

6. Future Professional Development Goals

The three goals below are sequenced on purpose. Each builds on capability established this semester, and each is scoped tightly enough to be assessable rather than aspirational. The first carries forward the deferred PLP Goal 3 (Software Development Tools) into a concrete deliverable; the second and third grow naturally out of the work completed this semester.

6.1 Goal 1 — Software Development Tools and Cloud Deployment (continuation of PLP Goal 3)

Rationale: I have made informal progress on Jira through group projects but have nothing concrete to show for the rest of PLP Goal 3 (CI/CD, Docker, NUnit), and LEXIS currently runs on localhost. Deploying LEXIS to AWS through a containerised CI/CD pipeline exercises all four of the original sub-skills in a single deliverable. Plan: within the next three months I will complete the AWS Certified Cloud Practitioner exam (estimated 40 hours of study), and deploy LEXIS to AWS using Docker, GitHub Actions for CI/CD, ECS Fargate, and RDS for PostgreSQL. The deliverable will be a live public URL, an architecture diagram, a green CI badge on the repository, and a passing exam score, all linked from the portfolio.

6.2 Goal 2 — Application-layer security and secure development lifecycle

Rationale: this semester's cybersecurity work stopped at the network layer. The complementary application-layer knowledge — OWASP Top 10, threat modelling, dependency scanning — is the obvious next step, and closes the loop on the original PLP Goal 2 scope (OWASP Foundation, 2021). Plan: over the next four months I will work through the OWASP WebGoat exercises locally, perform a documented STRIDE threat-modelling exercise on LEXIS, and integrate a dependency-vulnerability scanner (npm audit, Dependabot, or Snyk) into the CI pipeline established under Goal 6.1. The deliverable will be a written threat model and a continuously-passing security scan on the public repository.

6.3 Goal 3 — Responsible integration of AI APIs into production workflows

Rationale: generative AI shows up in most Australian software job ads, and Holocentric's product roadmap explicitly mentions AI-assisted compliance review. Demonstrating responsible LLM integration — cost, prompt injection, data handling — is a real differentiator at graduate level. Plan: within six months I will design and ship a document-summarisation feature inside LEXIS, calling either the Anthropic or OpenAI API from the Express back-end. The implementation will include a clear data-handling boundary (no client-identifying information leaves the system without explicit user action), structured logging of API costs, and basic prompt-injection mitigations. This goal explicitly closes the loop on Goal 1 from this semester: ethics stays a first-class architectural concern, not a postscript.

References

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